PORTFOLIO PROJECT

Student Performance and Behavior Analysis with Machine Learning

What 5,000 student records can and cannot tell a school about performance, stress and dropout risk

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THE PROBLEM

Schools want to know which students need support

If routine academic and lifestyle records could flag struggling students early, advisors could step in before grades fall or a student leaves. This project tests whether one such dataset can do that, through four tasks.

CLASSIFICATION

Fill missing labels

Infer parent education level for the 1,794 students with no record.

REGRESSION

Predict outcomes

Estimate each student's total score and stress level.

CLUSTERING

Find profiles

Group students into high-performing, average and struggling.

ANOMALY DETECTION

Flag dropout risk

Spot students whose records look unusual enough to warrant a check-in.

THE DATA

The dataset: 5,000 students, 23 attributes

5,000
student records

23
attributes per student

1,794
students with no parent education on record (36%). Attendance and assignment averages are each missing for about 10%.

Academic	Attendance, midterm, final, assignments, quizzes, participation, projects, total score, grade
Behavior and background	Study hours, sleep hours, stress level, extracurricular activities, internet access, parent education, family income
Demographics	Gender, age (18 to 24), department (CS, Engineering, Business, Mathematics)

Source: Students Grading Dataset, Kaggle

APPROACH

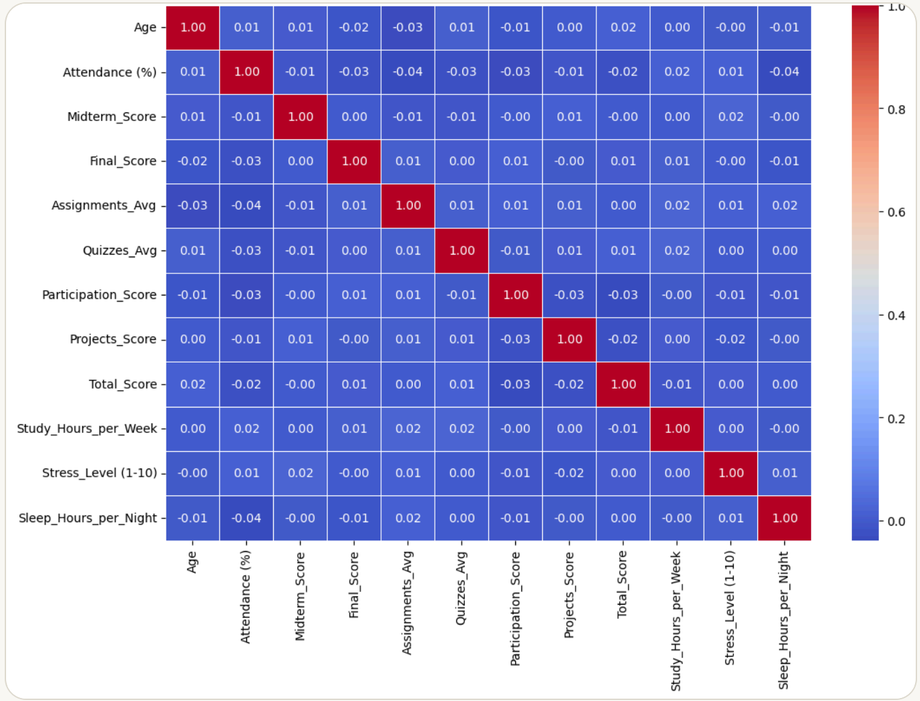
One pipeline feeds four machine learning tasks

Explore → Clean and impute → Engineer features → Tune 11 models → Evaluate

Task	Models compared	How they were judged
Classification	Random Forest, XGBoost, Gaussian Naive Bayes	10-fold stratified cross-validation accuracy
Regression	Linear, Polynomial (degree 1 to 4), Random Forest	RMSE and R^2 on a 20% hold-out
Clustering	K-Means, DBSCAN, Agglomerative	Silhouette score and visual inspection
Anomaly detection	Isolation Forest, Local Outlier Factor	Silhouette score

EXPLORATION

Exploration showed almost no relationship between features



0.04

Largest correlation between any two numeric features, in absolute value

Distributions look uniform. Scores, attendance, study hours, sleep and stress are spread almost evenly across their ranges, with no outliers in any boxplot.

A first warning sign. In real student data, grades usually move with attendance and study time. Here they do not, which suggests the dataset is synthetic.

Pearson correlation, numeric features

DATA PREPARATION

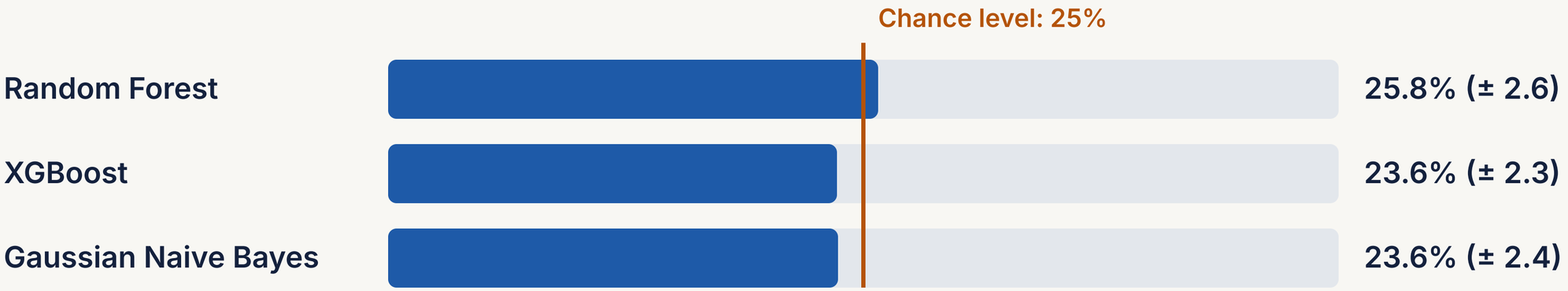
Each cleaning step was chosen to fit the column

Columns	Treatment	Reason
Assignment average (517 missing)	Department-wise mean	Keeps each department’s own average
Attendance (516 missing)	Global mean	Distribution is close to uniform
Parent education (1,794 missing)	Predicted by a classifier	Tested as an alternative to mode imputation
Participation score, sleep hours	Binned into low and high at 6	Turns a noisy scale into a simple flag
Gender, department, grade, income	One-hot encoding	Avoids implying an order between categories
Age, study hours per week	Standardization	Centers features measured in different units
Scores, attendance, stress level	Min-max scaling to 0 to 1	Puts all score-like features on one range

Identifier columns dropped before modeling

TASK 1 · CLASSIFICATION

No model predicts parent education better than chance



Mean accuracy over 10-fold stratified cross-validation, four balanced classes, bars on a 0 to 50% scale. Tree models tuned with RandomizedSearchCV.

What it means. Grades, attendance and lifestyle fields say nothing about parent education here. Filling the 1,794 missing labels with this classifier is close to assigning them at random.

EVALUATION CHECK

The hold-out set confirms the cross-validation result

Model	10-fold CV	Hold-out
Random Forest	25.8%	26.3%
XGBoost	23.6%	23.2%
Gaussian Naive Bayes	23.6%	24.5%

Hold-out accuracy is measured on 642 students the models never saw, with each model fit on the training split only. Every score is within two points of the 25% chance level.

A leak caught and fixed

An earlier run reported **92.2%** hold-out accuracy for Random Forest. That score came from a model fit during cross-validation, which had already seen about 90% of the hold-out rows.

$$0.9 \times 100\% + 0.1 \times 25\% \approx 92.5\%$$

Fix. The hold-out set is now scored only by the model fit on the training split.

TASK 2 · REGRESSION

Regression models could not beat predicting the average

Model	Total score RMSE	Total score R^2	Stress RMSE	Stress R^2
Linear Regression	0.2864	−0.0155	0.3131	−0.0023
Polynomial, degree 2	0.2966	−0.0892	0.3205	−0.0504
Random Forest, 400 trees	0.2957	−0.0287	0.3220	−0.0114

R^2 below zero

Every model explains less variance than a flat line at the mean.

Extra capacity hurts

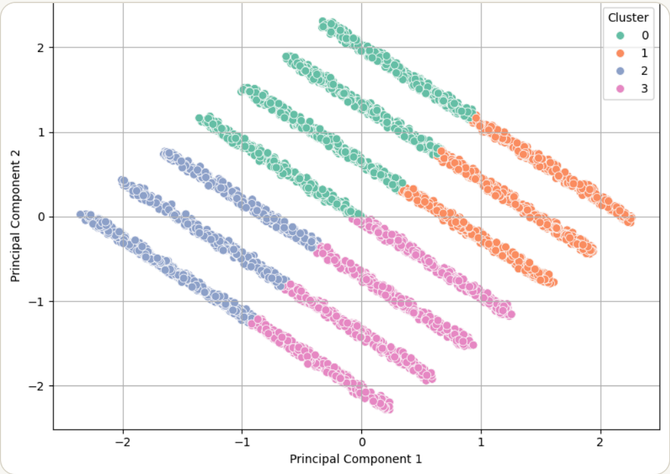
Polynomial error doubles at degree 3: 0.598 for total score, 0.740 for stress.

Simplest model wins

Linear Regression has the lowest error, by a margin too small to matter.

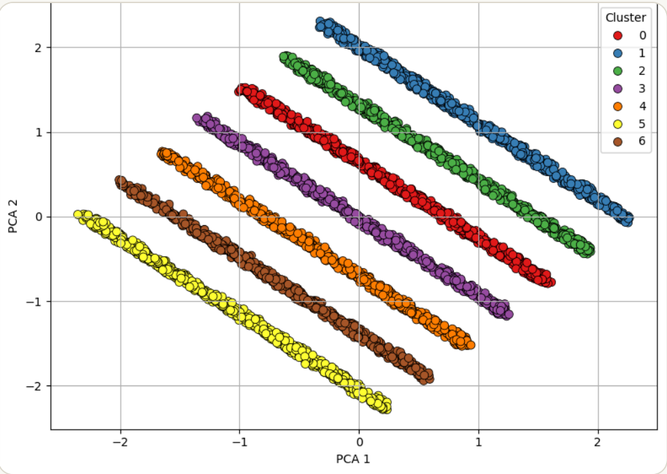
TASK 3 · CLUSTERING

The clusters follow age, not student performance



K-Means, k = 4

Silhouette 0.412. Clusters cut across the bands.



DBSCAN, 7 clusters

Silhouette 0.119. One cluster per band.

The bands are ages. The seven stripes match the seven ages in the data, 18 to 24. Only age and study hours were standardized, so they dominate both principal components.

Higher score, worse fit. K-Means and Agglomerative (k = 6, silhouette 0.364) score higher but split the bands.

No performance groups. No method found high-performing, average and struggling students.

PCA projection to two components

TASK 4 · ANOMALY DETECTION

Fifty students were flagged, but the flags are unvalidated

50 of 5,000

students labeled “potential dropout” by Isolation Forest, exactly 1%

Model	Best setting	Silhouette
Isolation Forest	300 trees, contamination 0.01	0.109
Local Outlier Factor	45 neighbors, contamination 0.01	0.102

Why these flags should not drive action

- The 1% share follows from the contamination setting chosen in tuning, not from the data.
- No dropout label exists, so no flag can be checked against what happened.
- Unusual is not the same as at risk. Where features carry no signal, outliers are arbitrary.

THE VERDICT

Four tasks and eleven models gave one answer: these records do not predict these outcomes.

A negative result is still a result. It tells a school not to build interventions on this data, and it shows the checks that keep a false success from reaching a decision-maker.

RECOMMENDATIONS

What a school should take from these results

01

Do not act on these predictions

Score forecasts, cluster labels and dropout flags from this dataset should not guide advising, funding or any decision about an individual student.

02

Collect outcome data first

Record actual dropouts and follow students over several terms. A model needs a known outcome to learn from and to be checked against.

03

Test for signal before modeling

A correlation check and a baseline model take minutes. They show whether a full modeling effort is worth the time before anyone commits to it.

NEXT STEPS

What we would change in the next iteration

Change	Why it matters
Wrap preprocessing and models in one scikit-learn Pipeline	Scalers and imputers are fit on training folds only, which removes leakage by design
Report a dummy baseline beside every model	Makes “no better than chance” visible on the first run
Measure feature and target dependence before tuning	Mutual information or a permutation test flags missing signal early
Use an “Unknown” category for missing parent education	A chance-level classifier only adds noise to 1,794 rows
Repeat the study on real institutional data	Shows whether the same methods work when signal exists

ABOUT THIS REPORT

Team, tools and source

TEAM

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Paulina Devina Wijaya
Reuben Zachary Susanto

TOOLS

Python, pandas, NumPy, scikit-learn, XGBoost, SciPy, Matplotlib, seaborn

CODE AND NOTEBOOK

github.com/ReubenZSTechs